There is another way of understanding the origin of the heat of black holes. The quantum fluctuations generate a correlation between the interior and the exterior of a hole. (I will speak at length about correlations and temperature in Chapter 12). Quantum uncertainty across the horizon of the black hole generates fluctuations of the horizon's geometry. But fluctuations imply probability, and probability implies thermodynamics, and therefore temperature. Concealing from us a part of the universe, a black hole makes its quantum fluctuations detectable in the form of heat.

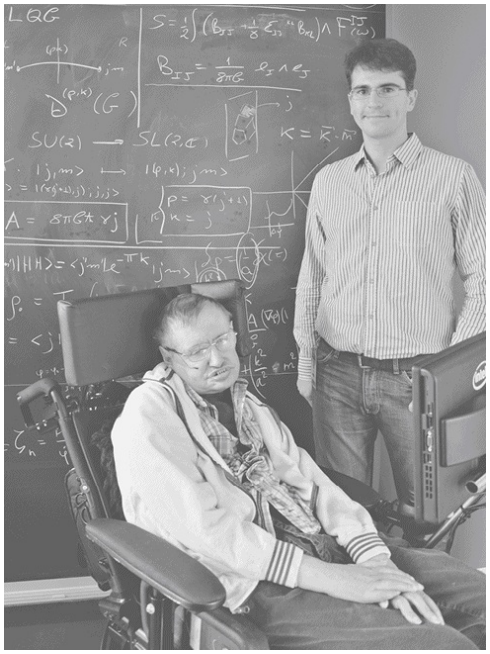


Figure 10.3 Stephen Hawking and Eugenio Bianchi. On the blackboard are the principal equations of loop quantum gravity which describe black holes.

It was a young Italian scientist, Eugenio Bianchi, today a professor in the United States, who completed an elegant calculation which shows how, starting from these ideas and from the basic equations of loop quantum gravity, it is possible to derive the formula for the heat of black holes foreseen by Hawking (figure 10.3).

The second application of loop quantum gravity to black-hole physics is more spectacular. Once collapsed, a star vanishes from external view: it is inside the black hole. But, inside the hole, what happens to it? What would you see if you let yourself fall into the hole?

At first, nothing in particular: you would cross the surface of the black hole without major injuries – then you would plummet towards the centre, at ever greater speed. And then? General relativity predicts that everything is squashed at the centre into an infinitely small point of infinite density. But this is, once again, if we ignore quantum theory.